

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 09)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

- A Radio collar ☐
- B Acoustic / satellite tag ☐
- C Pitfall trap ☐
- D Geolocator tag recovered by recapture ☐

0 1 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

5

0 2

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 2 . 1

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 2 . 2 Explain why herring populations recover from overfishing much faster than cod. **[2 marks]**

.....

.....

.....

0 2 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished. **[2 marks]**

.....

.....

.....

0 2 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species. **[2 marks]**

.....

.....

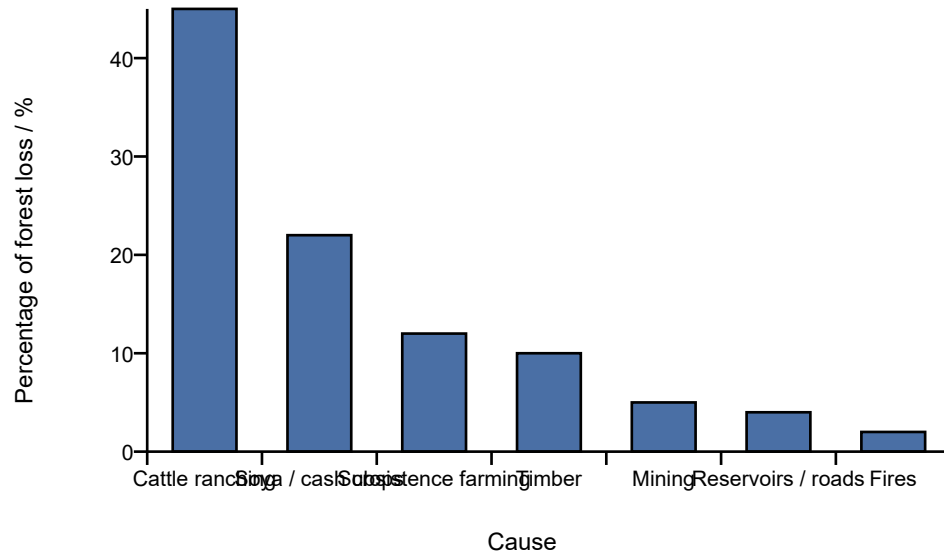
.....

10

03

Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



03

1

Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

03

2

Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

.....

.....

.....

.....

03

3

Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction

.....

HEP reservoirs

.....

Climate change

.....

0 3 . 4 Describe how a debt-for-nature swap works.

[2 marks]

.....

.....

.....

.....

10

0 4 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 4 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

.....

.....

.....

0 4 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

.....

04 . 3

Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer

Earths

04 . 4

Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

04 . 5

Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

1

2

10

0 5

Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha ⁻¹	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 5**. 1**

Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 5**. 2**

Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

05 . 3

Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

.....

.....

.....

.....

.....

.....

10

06 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

.....

Fuel

.....

Food

.....

Fibres

.....

Medicines

.....

06 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

.....

.....

.....

06 . 3

Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

10

07 . 1

Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Extra space

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

07 . 2 State what is meant by **permaculture**.

[1 mark]

.....

10

08 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

Extra space

08 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

10

09 . 1

Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

09 . 2

Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

09 . 3

Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

09 . 4

Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

10 . 1

Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

Extra space

10 . 2 State what is meant by **biodiversity**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.

END OF QUESTIONS